Courses Description

I. Mandatory Courses

Non-Major Courses

A description of the non-major mandatory courses follows.

ELEC 200	Introduction to Electrical and Computer Engineering	3(3,0)
In this course, the students will be trained in core skills to equip them to excel in electrical, biomedical, and computer and communications engineering. The student will be trained in application programming using an appropriate programming environment. Students will build on the basics of Programming learned in the introductory programming course to implement Digital Signal Processing applications and perform data analysis. Students will also be introduced to basic lab equipment such as Oscilloscopes, Function Generators, and Multi-Meters. Students will be introduced to microcontroller-based projects to further explore the Electrical and Computer Engineering majors (CCE, BIOM, and ELEC). Throughout the course, students will work in multidisciplinary teams to study engineering cases for the community and implement a project to address a community need.		

ELEC 210	Electric Circuits	3(3,0)
This course covers circuit elements and laws; mesh and node equations; network theorems; energy-storage elements; RC, RL, and RLC circuits; transformers; sinusoidal (AC) steady-state analysis; power calculation; and an introduction to three-phase circuits. **Prerequisite**: PHYS 191 or Equivalent.		

ELEC 320	Electronics	3(3,0)
This course covers semiconductors, PN junctions, diode theory and circuits, bipolar junction transistors, transistor fundamentals, transistor biasing, amplifiers, MOSFETs, and operational amplifiers. **Prerequisite:** ELEC 210.		

ELEC 320L	Electronics Lab	1(0,2)
This is a Lab course with experiments in Electrical and Electronic Circuits. It covers passive electrical elements and sources; lab instruments; voltage divider circuits; Thevenin's theorem; RC circuits; diode circuits; Op-Amp circuits; BJT and MOSFET characteristics. **Co-requisite:** ELEC 320.		

ELEC 340	Engineering Electromagnetics	3(3,0)
Introduction to vector analysis; Electrostatic fields in vacuum and dielectrics; Conductors; Capacitance; Electrostatic energy and forces; Poisson's equation. Magnetic fields, Maxwell's equations, static electric and magnetic fields, boundary-value problems, Laplace's and Poisson's equations, and static electromagnetic fields. Time-dependent Maxwell's equations and Plane-wave propagation. **Prerequisite**: PHYS 211.		

ENGL 210	English Composition and Rhetoric	3(3,0)
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, a research paper, and an oral presentation. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	Professional English Communication	3(3,0)
This required course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics; present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting and financial statements.		

ENGR 510	Engineering Project Management	3(3,0)
The course covers key components of engineering project management including projects election and planning, project time management, cost estimation and pricing, contract and specifications, quality management, engineering ethics and professional conduct, realizing impact of engineering solutions in various contexts (global, economic, environmental, societal, etc...), sustainability in engineering designs, human resources consideration, communications, risk management, and procurement management. **Prerequisite:** ENGR 300.		

MATH 210	Discrete Mathematics	3(3,0)
Logic, propositional equivalences, predicates and quantifiers, methods of proof, proof strategy, mathematical induction, recursive definitions, strong induction, and structural induction, sets and set operations, functions, growth of functions, relations, and their properties, representing relations, equivalence relations, introduction to graphs, and graph terminology.		

MATH 215	Calculus III	3(3,0)
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar		

curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series. **Prerequisite:** MATH 215.		

MATH 311	Linear Algebra with Applications	3(3,0)
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	Ordinary Differential Equations	3(3,0)
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

MATH 421	Numerical Analysis	3(3,0)
Error Analysis, solutions of nonlinear equations using fixed-point, Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the above-mentioned topics. **Prerequisite**: MATH 311.		

PHYS 211	Physics: Electricity and Magnetism and Lab	3(3,1)
This course covers fundamental topics in Electricity and Magnetism: Electric forces and Electric Fields for discrete and continuous charge distributions; Gauss's Law; Electric Potential; Capacitance and Dielectrics; Kirchhoff's rules; Magnetic Fields and Forces; Biot-Savart Law; Ampere's Law; Magnetic Flux and Gauss's Law in Magnetism. It also includes a laboratory component that introduces students to the "real world" through experiments in electricity and magnetism. **Prerequisite**: None.		

Major Courses

Mandatory computer and communications engineering courses are described below.

CCEE 214	Introduction to Programming	3(2,2)
This course presents the fundamentals of structured programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to COSC 214.		

CCEE 216	Advanced Programming and Data Structures	3(2,2)
This course covers advanced object-oriented programming concepts, including overloading, inheritance, and polymorphism. In addition, the course covers data structure concepts, including analysis, sorting, and searching algorithms, as well as stacks, queues, trees, and graphs. **Prerequisite**: CCEE 214. Equivalent to COSC 215.		

CCEE 221	Logic Design	3(3,0)
This course addresses Boolean algebra and logic simplification techniques, data representation, and the design of combinational logic networks for decoders, encoders, multiplexers, and demultiplexers; the design of sequential logic devices, including flip-flops, registers, and counters, as well as the analysis of devices used to build logic networks, including programmable logic devices. **Equivalent to** COSC 351.		

CCEE 221L	Logic Design Lab	1(0,2)
This Lab covers design techniques and implementation of combinational and sequential logic circuits. Experiments include logic gates; the design and implementation of logic circuits; combinational logic circuits (decoders, encoders, multiplexers, demultiplexers, and adders); and the design of sequential logic devices using flip-flops, registers, and counters. **Co-requisite**: CCEE 221. **Equivalent to** COSC 351L.		

CCEE 310	Software Engineering	3(3,0)
Analysis, design, construction, maintenance, and evolution of large software systems are covered. Students are introduced to the system life cycle, project management techniques, and database systems. Analysis, design, and implementation of software systems are also included. **Prerequisite**: CCEE 214. **Equivalent to** COSC 341.		

CCEE 315	Database Management Systems	3(3,0)
The course covers the nature and purpose of database systems; an introduction to data modeling: the Entity-Relationship Model and the Relational Model, including relational algebra, relational calculus, and SQL; integrity constraints; file organization and index files; and normalization. It also covers an introduction to object databases, data mining, schema evolution, distributed databases, web-enabled databases, and databases for e-commerce applications. **Prerequisite**: CCEE 214. **Equivalent to** COSC 231.		
CCEE 324	**Computer Organization**	**3(3,0)**

Computer system organization and design; implementation of CPU data path and control; instruction set design; memory hierarchy (caches, main memory, virtual memory) organization and management; input/output subsystems (bus structures, interrupts, DMA); performance evaluation; and pipelined processors. **Prerequisite**: CCEE 221. **Equivalent to** COSC 353.

CCEE 331	**Signals and Systems**	**3(3,0)**
This course covers the mathematical description and classification of continuous and discrete signals and systems. Topics include types of signals and systems, Fourier series, Fourier transforms, Discrete-Time Fourier Transforms (DTFT), Discrete and Fast Fourier Transforms (DFT and FFT), Laplace Transforms, Z-Transforms, and transfer functions. **Prerequisite:** MATH 216		

CCEE 341	**Communication Systems**	**3(3,0)**
This course introduces the fundamentals of transmission and reception in communication systems, as well as the effects of noise. Topics include power spectral density; amplitude modulation and demodulation; angle modulation and demodulation; analog communication system performance in the presence of noise; sampling and analog-to-digital conversion; and an introduction to digital modulation. **Prerequisite**: CCEE 331.		

CCEE 354	**Computer Networks**	**3(3,0)**
This course enables students to gain fundamental knowledge of computer networks, appreciate the trade-offs and choices in networking, learn to design and analyze protocols, and prepare to study advanced topics in the field. **Co-requisite**: CCEE 221. Equivalent to COSC 360.		

CCEE 354L	**Computer Networks Lab I**	**1(0,2)**
This Lab helps prepare students seeking to pass advanced Cisco Certifications. The student will acquire knowledge of network equipment and protocol functionality, learn to build a simple Ethernet network using routers, switches, and computers, and use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts. **Co-requisite**: CCEE 354.		

CCEE 400	**Summative Learning (BS) Project**	**3(3,0)**
Team-oriented, project-based experience culminating in the creation of an artifact; milestones include project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the device. **Prerequisite**: ENGL 217 taken in the last semester of the BS Program.		

CCEE 411	**Web Programming**	**3(3,0)**
This course teaches students how to develop and implement web-based programs with an emphasis on interface programming. It introduces students to web development and the client-side and server-side languages and technologies needed to build functional, responsive websites. The course covers HTML5, CSS3, JavaScript/jQuery, PHP, and responsive design. **Co-requisite:** CCEE 214. Equivalent to COSC 333.		

CCEE 411L	Internet Engineering and Web Programming Lab	1(0,2)
This lab provides students with hands-on skills in Web programming. Experiments mainly cover the following topics: HTML and CSS, JavaScript and JQuery, Responsive web design, Bootstrap, Web content management systems, and server-side development. **Co-requisite:** CCEE 411.		

CCEE 426	Design of Embedded Systems	3(3,0)
This course addresses the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, operating systems and scheduling, system-level simulation, and interface-based design. **Prerequisite**: CCEE 221. Equivalent to COSC 356.		

CCEE 426L	Design of Embedded Systems Lab	1(0,2)
This lab complements topics covered in the CCEE 426 course. It involves the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Experiments include Design using finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, scheduling, system-level simulation, and interface-based design. **Co-requisite**: CCEE 426. Equivalent to COSC 356L.		

CCEE 444	Antennas and Propagation	3(3,0)
This course introduces the fundamental principles of antennas and propagation for communications-oriented electrical engineers. Topics include: fundamental antenna parameters, wire antennas, analysis and design of linear antenna arrays, radio wave propagation, free-space path loss, and link budget. **Prerequisite**: ELEC 340.		

CCEE 444L	Antennas and Propagation Lab	1(0,2)
This lab provides students with hands-on experience in simulating and designing basic antennas using a microwave computational tool. It also includes a demonstration of basic antenna performance measurements. **Co-req.**: CCEE 444.		

CCEE 447	Digital Communications	3(3,0)
This course introduces the fundamentals of modern digital communication systems. Topics include: deterministic and random signal analysis; Channels; signal space representation; Maximum Likelihood Detection; Coherent and Non-Coherent Digital Modulation Schemes (ASK, PSK and M-PSK, FSK and M-FSK, M-QAM); error probabilities; bandwidth efficiency and energy efficiency tradeoffs; inter-symbol interference (ISI) and pulse shaping; matched filter and optimal receiver; source coding. **Prerequisite**: CCEE 341.		

CCEE 447L	Communications Lab	1(0,2)
This lab taken with CCEE 447. The student will acquire knowledge on implementing and analyzing analog and digital communication and systems using the NI EMONA Add-in Module and the NI USRP. analog and digital modulation, sampling and reconstruction, coding, and **Co-requisite:** CCEE 447.		

CCEE 449	Wireless Communications	3(3,0)
Wireless communications are omnipresent in today's world. It does not transport only conversational information; it is used for a plethora of tasks, from short-range exchange of photos over Bluetooth to satellite communication and deep-space information transfer, from a few bits used to control robots and for sensory information to big data collection. This course introduces current wireless systems and standards, shadowing and multipath fading effects in wireless communications, radio channel capacity, multiple access techniques, and channel coding methods. **Co-requisite:** CCEE 447.		

CCEE 460	Artificial Intelligence	3(3,0)
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include intelligent agents, informed and uninformed search strategies, and adversarial search. **Prerequisite**: CCEE 214 or equivalent.		

CCEE 460L	Artificial Intelligence Lab	1(0,2)
This lab will help prepare students to handle the huge amount of data produced by both humans and machines today. The student will learn to use state-of-the-art AI tools to analyze data and make complex decisions based on that data using real-world practical examples and case studies. At the end of the lab, the student will be able to apply AI knowledge in their field at work or in day-to-day life to make better decisions. **Co-req.:** CCEE 460.		

CCEE 499	Co-op Work Experience	3(3,0)
This Co-op work experience is designed to provide students with an eight-week short-term work experience in the field of computer and communication engineering. Students are encouraged to network with industry professionals in their discipline. A report, poster, and PowerPoint presentation are required. **Prerequisites**: Senior Standing. ENGL 217.		

CCEE 528	Computer Architecture	3(3,0)
Computer system organization and design; implementation of CPU data path and control; instruction set design; memory hierarchy (caches, main memory, virtual memory) organization and management; input/output subsystems (bus structures, interrupts, DMA); performance evaluation; pipelined processors. **Prerequisite**: CCEE 324. **Equivalent to** COSC 353.		

CCEE 534	Signal Processing	3(3,0)
This course covers time series analysis, frequency analysis, time-frequency analysis, and time-scale analysis. It also covers the design of digital filters and signal modeling. **Prerequisite**: CCEE 331.		